

AI STUDYBUDDY API

Solution Architecture

Application layers and key request flows

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TEAM SIZE	5
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Layers

Client -> Express entry point -> route groups -> auth/admin/upload middleware -> controllers -> Mongoose models -> MongoDB. Material AI calls go through Gemini helper to Google Generative AI.

Authentication flow

Register/login -> User lookup/save and bcrypt -> token utility signs access token (15m) and refresh token (7d). Protected middleware verifies access token from Bearer header.

Material flow

Multer saves accepted file to uploads/ -> controller reads as UTF-8 -> Material record persists -> temp file is removed. Listing filters by student owner; admin listing can see all.

AI flow

Fetch material -> create task-specific Gemini prompt -> parse generated JSON where flashcards/quiz are expected -> save output on Material.

Reliability/security risks

Malformed model output can fail JSON parsing. AI endpoints do not visibly perform owner checks. Public registration accepts role. Treat these as code review findings, not verified exploit tests.